



Islamic University of Technology  
Department of Computer Science and Engineering



# Team System322 Project Report

## Lab 10- Process Specifications and Structured Decisions

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## Abstract

In response to consistent communication challenges between students and the administration at Islamic University of Technology (IUT), our project has aimed to develop a structured, reliable platform to support transparent dialogue and issue reporting. The selected organization, the IUT Office of Students' Welfare (IUT-OSW), currently serves as the official bridge between students and the administration but lacks a reliable digital platform to support open dialogue and issue reporting, which in turn causes student concerns often being voiced through informal, unmoderated platforms like Facebook. Labs 1 through 9 have progressively shaped our solution—IUTian Threads—a university-hosted, Reddit-style discussion forum designed to empower student voices while ensuring accountability and constructive engagement. Through feasibility analysis, stakeholder research, and iterative requirement gathering and prototyping, we validated the need for a secure, anonymous, and moderated platform aligned with OSW's mission and student expectations.

In Lab 10, we took a look at our process specification and structured decisions. While our DFDs showed the data flow in the whole system here we detail the specifics of our processes that were presented in lab 8. Among our processes in the Diagram 0 and subsequent child diagrams, we have selected 3 processes for detailed showcase and applied structured decision techniques such as decision tables, Structured English and Decision Tree. Additionally we documented the specifics like input and output flows in the subsequent process forms and added any ambiguity and unresolved issues we have faced.

## 1. Candidate Process and Selection

### **List of Candidate Primitive Processes**

From our Level 0 and Child Diagrams developed in Lab 8, we identified the following processes:

#### Level 0 Processes (Main Processes):

1. **Process 1 - Login Page:** Handles user authentication and session management
2. **Process 2 - Post Interaction Handler:** Manages all post-related interactions (decomposes further)
3. **Process 3 - Subforum Process:** Handles subforum search and navigation
4. **Process 4 - Feed Generator:** Creates personalized content feeds for users
5. **Process 5 - Moderation Toolset:** Manages content moderation and approval workflows
6. **Process 6 - Notification Service:** Distributes alerts and notifications to users

#### Child Processes (Primitive Processes from Process 2 decomposition):

1. **Process 2.1 - Store Comment:** Processes and stores user comments with validation and threading
2. **Process 2.2 - Register Vote:** Handles upvote/downvote actions with duplicate prevention
3. **Process 2.3 - Post Update Relay:** Manages post modification notifications and updates
4. **Process 2.4 - Posting & Anonymous Posting:** Creates new posts with content validation and moderation alerts

## **Selected Processes for Specification**

We have selected three processes that represent distinct types of decision logic and business rules:

### **i) Process 2.4 - Posting & Anonymous Posting**

**Justification:** This process embodies critical business rules around user authentication, content validation, and anonymity handling. It demonstrates complex conditional logic where multiple factors (authentication status, post type, content validity) interact to determine appropriate actions. The process is central to the platform's core functionality and requires precise specification to ensure consistent handling of both regular and anonymous posts while maintaining moderation workflows.

### **ii) Process 2.1 - Store Comment**

**Justification:** This process represents sequential validation logic with nested conditional checks. It demonstrates how the system handles hierarchical validation (authentication → post existence → content validation) with clear error handling at each stage. The process is complex enough to benefit from formal specification while being fundamental to user engagement and content interaction within the platform.

### **iii) Process 2.2 - Register Vote**

**Justification:** This process involves asymmetric decision trees with multiple branching paths based on existing vote status and user actions. It handles duplicate vote prevention, vote changes, and score updates through cascading decisions that create natural tree-like logic flow. The process is crucial for maintaining data integrity in the voting system and preventing manipulation.

## 2. Process Specification Forms & Analysis

### Process Specification Form 1- Posting & Anonymous Posting

|                                                                                                                                                                 |                                                                                                     |                                                                                                                            |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--|
| <b>Number</b>                                                                                                                                                   | 2.4                                                                                                 |                                                                                                                            |  |
| <b>Name</b>                                                                                                                                                     | Posting & Anonymous Posting                                                                         |                                                                                                                            |  |
| <b>Description</b>                                                                                                                                              | Handles creation of new posts including anonymous posts with content validation & moderation alerts |                                                                                                                            |  |
| <b>Input Data Flow</b>                                                                                                                                          |                                                                                                     | <b>Output Data Flow:</b>                                                                                                   |  |
| <ul style="list-style-type: none"> <li>New post content from student, user authentication token</li> </ul>                                                      |                                                                                                     | <ul style="list-style-type: none"> <li>Stored post to In-Subforum Content Master, moderation alert to moderator</li> </ul> |  |
| <b>Type of Process:</b> <input checked="" type="checkbox"/> Online <input type="checkbox"/> Batch <input type="checkbox"/> Manual                               |                                                                                                     |                                                                                                                            |  |
| <b>Decision Analysis:</b> <input type="checkbox"/> Structured English <input checked="" type="checkbox"/> Decision Table <input type="checkbox"/> Decision Tree |                                                                                                     |                                                                                                                            |  |

Image 1: Process specification form of Process 2.4

### Decision Table

| Conditions & Actions              | Rule 1 | Rule 2 | Rule 3 | Rule 4 |
|-----------------------------------|--------|--------|--------|--------|
| User is Authenticated?            | Y      | Y      | Y      | Y      |
| Post type=Anonymous?              | Y      | N      | Y      | N      |
| Content Passes Validation?        | Y      | Y      | N      | N      |
| <b>Actions</b>                    |        |        |        |        |
| Store post with hidden author ID  | X      |        |        |        |
| Store post with visible author ID |        | X      |        |        |
| Reject post submission            |        |        | X      | X      |
| Send Moderation alert             | X      |        | X      | X      |

Table 1: Decision table of Process 2.4

## Process Specification Form 2 - Store Comment

|                           |                                                                                                                                                                                                                                                          |  |  |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| <b>Number</b>             | 2.1                                                                                                                                                                                                                                                      |  |  |
| <b>Name</b>               | Store Comment                                                                                                                                                                                                                                            |  |  |
| <b>Description</b>        | Processes and stores user comments (and replies) on posts with validation, threading, and moderation cues.                                                                                                                                               |  |  |
| <b>Input Data Flow</b>    | <b>Output Data Flow:</b> <ul style="list-style-type: none"> <li>Updated <i>In_Subforum_Content_Master</i> (append comment_record, increment comment_count)</li> <li>success_confirmation</li> <li>moderation_alert to <i>Moderation_Queue</i></li> </ul> |  |  |
| <b>Type of Process:</b>   | <input checked="" type="checkbox"/> Online <input type="checkbox"/> Batch <input type="checkbox"/> Manual                                                                                                                                                |  |  |
| <b>Decision Analysis:</b> | <input checked="" type="checkbox"/> Structured English <input type="checkbox"/> Decision Table <input type="checkbox"/> Decision Tree                                                                                                                    |  |  |

Image 2: Process specification form of Process 2.1

## Structured English

```

BEGIN Store_Comment
  RECEIVE comment_content, post_id, user_auth FROM
  input_flow
  IF user_auth IS valid THEN
    IF post_id EXISTS IN In_Subforum_Content_Master THEN
      VALIDATE comment_content FOR prohibited_words
      IF validation PASSES THEN
        CREATE comment_record WITH timestamp
        APPEND comment_record TO post.comments_array
        UPDATE post.comment_count
        SEND success_confirmation TO user
      ELSE
        SEND validation_error TO user
      ENDIF
    ELSE
      SEND post_not_found_error TO user
    ENDIF
  ELSE
    SEND authentication_error TO user
  ENDIF
END Store_Comment

```

## Process Specification Form 3 - Register Vote

|                                                                                                                                                                      |                                                                                  |                                                                                                                             |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|--|
| <b>Number</b>                                                                                                                                                        | 2.2                                                                              |                                                                                                                             |  |
| <b>Name</b>                                                                                                                                                          | Register Vote                                                                    |                                                                                                                             |  |
| <b>Description</b>                                                                                                                                                   | Processes upvotes and downvotes on posts/comments with duplicate vote prevention |                                                                                                                             |  |
| <b>Input Data Flow</b>                                                                                                                                               |                                                                                  | <b>Output Data Flow:</b>                                                                                                    |  |
| <ul style="list-style-type: none"> <li>• Vote Action (upvote/downvote)</li> <li>• target_post_ID of the post interacted with</li> <li>• UUID of the user.</li> </ul> |                                                                                  | <ul style="list-style-type: none"> <li>• Updated vote count in In-Subforum Content Master</li> <li>• Post update</li> </ul> |  |
| <b>Type of Process:</b> <input checked="" type="checkbox"/> Online <input type="checkbox"/> Batch <input type="checkbox"/> Manual                                    |                                                                                  |                                                                                                                             |  |
| <b>Decision Analysis:</b> <input type="checkbox"/> Structured English <input type="checkbox"/> Decision Table <input checked="" type="checkbox"/> Decision Tree      |                                                                                  |                                                                                                                             |  |

Image 3: Process specification form of Process 2.2

## Decision Tree:

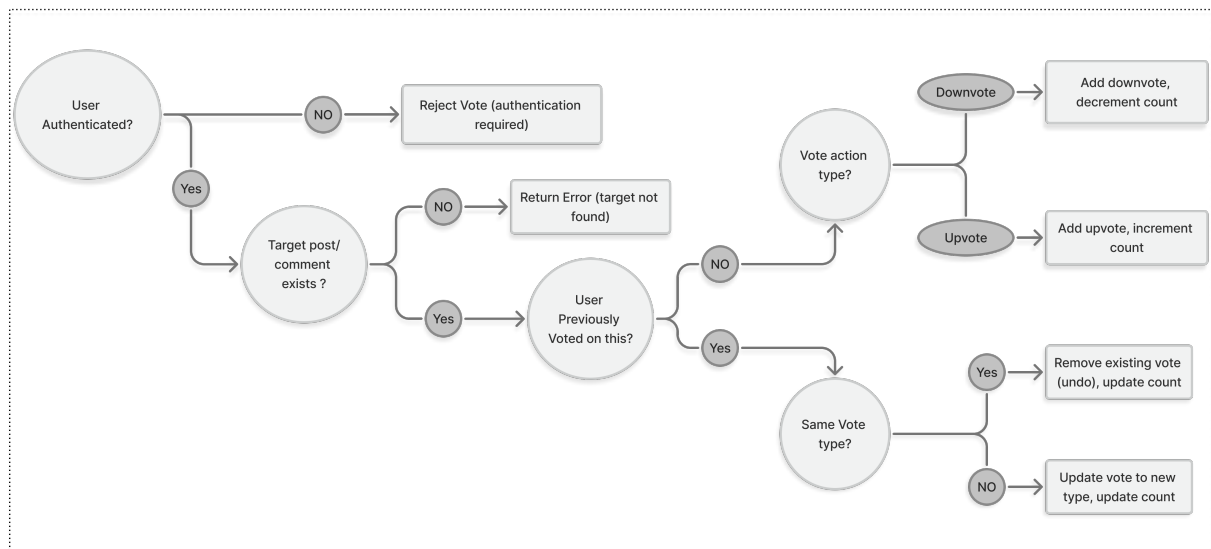


Image 4: Decision Tree of Process 2.2

## Logic & Justification of Selected Decision Analysis

### Process 2.4 - Posting & Anonymous Posting (Decision Table)

The Decision Table technique is ideal for Process 2.4 because:

- i) Multiple Interacting Conditions:** The system validates user authentication, post type, and content validation, all of which interact to determine the appropriate action.
- ii) Clear Rule Mapping:** Each combination of conditions maps to specific actions, avoiding ambiguity in post handling.
- iii) Systematic Coverage:** The table ensures all possible combinations of conditions are addressed.

The system validates user authentication and post type to determine the storage method. Anonymous posts hide author identity while maintaining accountability through session tracking.

### Process 2.1 - Store Comment (Structured English)

Structured English is the optimal technique for Process 2.1 because:

- i) Sequential Flow:** Logic follows a step-by-step validation order that maps naturally to structured English syntax.
- ii) Nested Conditions:** Each check depends on the previous one (authentication → post existence → content validation), creating clear hierarchical logic.
- iii) Clear Error Handling:** Every failure point sends a meaningful error message; the successful path updates the database and confirms the action.

The process receives comment data and systematically validates each requirement before storing the comment, ensuring data integrity and providing meaningful feedback for all outcomes.

### Process 2.2 - Register Vote (Decision Tree)

The Decision Tree technique is most suitable for Process 2.2 because:

- i) Branching Decisions:** Vote processing involves multiple decision points where each choice determines the next step in the process.



**ii) Asymmetric Logic:** The decision paths are not uniform—some branches lead to immediate actions while others continue deeper into the logic.

**iii) Clear Flow Visualization:** The tree structure makes the duplicate vote handling and vote change logic immediately apparent.

The tree clearly shows how the system handles existing votes, prevents duplicates, and manages vote changes while maintaining referential integrity in the voting system.

### 3. Ambiguities and Unresolved Issues

During the process of specifying these critical system processes, several challenges and ambiguities emerged that require careful consideration:

#### **Anonymous Post Traceability**

**Challenge:** Balancing user anonymity with moderation accountability presents a complex design challenge. While it is out focus that anonymous posts must hide user identity from other students, the system requires some mechanism to track posts for abuse prevention and policy enforcement.

**Current Approach:** Store posts with hidden author IDs while maintaining session-based tracking for moderation purposes. However, questions remain about long-term accountability and repeat offender identification without compromising anonymity principles.

## **Content Validation Rules**

**Challenge:** Defining comprehensive prohibited content filters without over-censoring legitimate academic discussions requires nuanced rule development. The system must be able to distinguish between constructive criticism of university policies and inappropriate content.

**Consideration:** Implementation of tiered validation with keyword filtering, context analysis, and human moderator review for edge cases. The balance between automated filtering and human judgment needs further refinement.

## **Vote Manipulation Prevention**

**Challenge:** Preventing coordinated voting attacks or bot-driven vote manipulation in the university environment while maintaining system performance and user experience.

**Approach:** Current specification includes duplicate vote prevention and user authentication requirements, but sophisticated manipulation schemes (multiple accounts, coordinated timing) require additional detection mechanisms.

## **Concurrent Operations**

**Challenge:** Handling simultaneous votes, comments, or posts on the same content without creating data conflicts or inconsistent states.

**Technical Consideration:** Proper database implementation and transaction management need to be specified to prevent race conditions while maintaining system responsiveness.

## **Moderation Escalation**

**Challenge:** Determining when content requires immediate moderator attention versus standard review queue processing lacks clear criteria in current specifications.

**Resolution Needed:** Development of automated flagging criteria based on content analysis, user reporting frequency, and severity indicators to optimize moderator workload distribution.

This report thus highlighted the complexity hidden within seemingly straightforward processes and demonstrated how formal specification prevents ambiguity while ensuring comprehensive coverage of business rules and edge cases.

— End of Report —